# Electrical load list

Flash drum D-400 unit · load list, maximum demand, UPS, cable schedule, hazardous-area classification, loss of power

## Document no. FD-V1-EE-LL-001

Every electrical consumer of the unit with its supply, board, duty, rated and absorbed power; the maximum demand and the UPS; the cable schedule sized for current and voltage drop; the hazardous-area classification with the gas group and temperature class; and what every fail-safe element does on loss of power, checked on a copy of the simulator model.

### Contents

1 Basis
2 Load list
3 Maximum demand and UPS
4 Cable schedule
5 Hazardous-area classification
6 Loss of power
7 Engineering queries

### Holds and engineering queries

- EE-01 Analyser shelter not on the plot plan
- EE-02 P-400 keeps running against a closed XV-408
- EE-03 SIS UPS autonomy not specified
- EE-04 AT-400 sample line also needs tracing
- EE-05 No gas detection in the unit

**Note.** Numbers come from the simulator model (design case and steady model runs) and the stated assumptions; generated for teaching, not for construction. Engineering queries raised by this document are listed in it and in FD-V1-PM-DOC-001.

| Rev | Date | Description | Originator | Checked | Approved |
|---|---|---|---|---|---|
| 0 | 28-OCT-26 | Issued for teaching | A. Soottitantawat | — | — |

| DOCUMENT NO. | REV | STATUS | ORIGINATOR | DATE |
|---|---|---|---|---|
| FD-V1-EE-LL-001 | 0 | Issued for teaching | Apinan Soottitantawat | 28-OCT-26 |

# 1 · Basis

| | |
|---|---|
| Scope | Unit D-400 / E-400 with its instruments, SIS, analysers, tracing, lighting and sockets. P-400 (feed pump) belongs to the upstream unit and is shown as an interface load only. No motor and no motor-operated valve in this unit. |
| Supply | 400 / 230 V, 50 Hz from DB-400 in the substation (assumed); 230 V AC UPS in the control building for the DCS and the SIS |
| Diversity | continuous 1.0, night lighting 1.0, intermittent (sockets) 0.3, standby 0 (typical) |
| Cables | Cu / XLPE 90 °C multicore on perforated tray (IEC 60364-5-52 method E, typical ratings); ambient 40 °C (assumed design maximum); ΔU running ≤ 5 %, motor start ≤ 15 % at 6.5 × FLC, pf 0.3; minimum 2.5 mm²; length = 1.15 × (\|Δx\| + \|Δy\|) + 10 m from the plot plan |
| Hazardous areas | IEC 60079-10-1 / API RP 505 method with typical extents: Zone 2 3 m around secondary sources, Zone 1 1 m around primary sources, Zone 2 band at grade to 7.5 m, 0.6 m high (vapour heavier than air) |

## 2 · Load list

| Tag | Consumer | Supply | Board | Duty | Rated kW | Absorbed kW | pf | UPS | Scope | Basis |
|---|---|---|---|---|---|---|---|---|---|---|
| **P-400** | Feed supply pump motor (upstream unit, interface only) | 400 V 3-ph | Upstream MCC | Continuous | 7.50 | 4.04 | 0.84 | — | Upstream | absorbed 4.04 kW at design, 4.77 kW at the maximum rate (η 55 %, assumed); 7.5 kW motor (Mechanical tab) |
| **DCS-400** | DCS controller share, I/O cards, marshalling and field loop power (23 installed I/O) | 230 V AC UPS | UPS-DB (control building) | Continuous | 0.38 | 0.32 | 0.90 | yes | This unit | 23 I/O × 10 W + 150 W controller / network share (typical); from the I/O count of IN-INL-001 |
| **SIS-400** | SIS logic-solver share, I/O cards and the XV solenoids (16 installed I/O) | 230 V AC UPS / 24 V DC | UPS-DB (control building) | Continuous | 0.34 | 0.30 | 0.90 | yes | This unit | 16 I/O × 10 W + 150 W logic solver share + 3 solenoids × 11 W, energised all the time (de-energise to trip) |
| **AN-410** | Process GC for AT-410 (bottoms, liquid sample) | 230 V | ASH-DB | Continuous | 0.80 | 0.60 | 0.95 | — | This unit | typical process GC (oven, electronics) |
| **AN-420** | Process GC for AT-420 (vapour) | 230 V | ASH-DB | Continuous | 0.80 | 0.60 | 0.95 | — | This unit | typical process GC |
| **AN-400** | $N_2$ analyser AT-400 (thermal conductivity) | 230 V | ASH-DB | Continuous | 0.20 | 0.15 | 0.95 | — | This unit | typical |
| **SC-410** | AT-410 sample conditioning: heated vaporising regulator | 230 V | ASH-DB | Continuous | 0.30 | 0.20 | 1.00 | — | This unit | typical |
| **HT-420** | Heat tracing of the AT-420 vapour sample line (dew point 49.2 °C) | 230 V | ASH-DB | Continuous | 0.45 | 0.36 | 1.00 | — | This unit | 15 m × 30 W/m self-regulating (assumed); EP-09 |
| **HT-400** | Heat tracing of the AT-400 vapour sample line | 230 V | ASH-DB | Continuous | 0.45 | 0.36 | 1.00 | — | This unit | same vapour as AT-420: condenses below 49 °C (EE-04) |
| **ASH-400** | Analyser shelter HVAC (pressurisation fan, cooling) and lighting | 400 V 3-ph | ASH-DB | Continuous | 3.00 | 2.40 | 0.85 | — | This unit | one shelter for three analysers (typical) |
| **LT-A** | Area lighting of the unit plot 32 × 24 m | 230 V | DB-400 | Night | 1.20 | 1.20 | 0.95 | — | This unit | 8 × 150 W LED floodlights, Ex ec (typical) |
| **LT-P** | D-400 platform, ladder and valve-station lighting | 230 V | DB-400 | Night | 0.24 | 0.24 | 0.95 | — | This unit | 6 × 40 W LED (typical) |
| **LT-E** | Escape lighting (self-contained batteries, 3 h) | 230 V | DB-400 | Standby | 0.20 | 0.02 | 0.95 | — | This unit | charging load only in normal service (typical) |
| **SO-400** | Welding / maintenance socket outlet 63 A, 400 V | 400 V 3-ph | DB-400 | Intermittent | 25.00 | 25.00 | 0.85 | — | This unit | typical welding set; socket and cable rated 63 A |
| **SO-401** | Convenience socket outlets 4 × 16 A, 230 V | 230 V | DB-400 | Intermittent | 2.00 | 2.00 | 0.90 | — | This unit | hand tools (typical) |

## 3 · Maximum demand and UPS

| Load | Diversity | Demand kW | kVA |
|---|---|---|---|
| DCS-400 | 1.00 | 0.323 | 0.359 |
| SIS-400 | 1.00 | 0.296 | 0.329 |
| AN-410 | 1.00 | 0.600 | 0.632 |
| AN-420 | 1.00 | 0.600 | 0.632 |
| AN-400 | 1.00 | 0.150 | 0.158 |
| SC-410 | 1.00 | 0.200 | 0.200 |
| HT-420 | 1.00 | 0.360 | 0.360 |
| HT-400 | 1.00 | 0.360 | 0.360 |
| ASH-400 | 1.00 | 2.400 | 2.824 |
| LT-A | 1.00 | 1.200 | 1.263 |
| LT-P | 1.00 | 0.240 | 0.253 |
| LT-E | 0.00 | 0.000 | 0.000 |
| SO-400 | 0.30 | 7.500 | 8.824 |
| SO-401 | 0.30 | 0.600 | 0.667 |
| **Maximum demand** | | **14.83** | **16.68** (pf 0.89) |

| | |
|---|---|
| UPS loads | DCS-400, SIS-400: 0.723 kW = 0.803 kVA (pf 0.9) |
| UPS rating | 2 kVA (≥ 1.25 × load) |
| Battery | 0.50 kWh for 30 min — autonomy 30 min (assumed), inverter η 0.9, battery depth of discharge 0.8 (typical) |

## 4 · Cable schedule

| Feeder | From | To | kW | V / ph | L m | I A | $k_T \cdot k_G$ | Size mm² | $I_z$ A | ΔU run % | ΔU start % | Governed by |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **F-01** | Upstream MCC (shown for teaching) | P-400 motor | 7.50 | 400 / 3 | 84 | 14.6 | 0.746 | 2.5 | 23.9 | 4.25 | 10.1 | running voltage drop |
| **F-02** | DB-400 (substation) | ASH-DB, analyser shelter | 6.00 | 400 / 3 | 59 | 9.6 | 0.746 | 2.5 | 23.9 | 2.10 | — | minimum size |
| **F-03** | DB-400 (substation) | SO-400 welding socket 63 A (sized for the 63 A socket rating) | 37.10 | 400 / 3 | 68 | 63.0 | 0.746 | 16.0 | 74.6 | 2.39 | — | current |
| **F-04** | DB-400 (substation) | LT-A area lighting circuit (8 floodlights) (+ 40 m around the plot (assumed)) | 1.20 | 230 / 1 | 102 | 5.5 | 0.746 | 2.5 | 26.9 | 4.38 | — | running voltage drop |
| **F-05** | DB-400 (substation) | LT-P platform lighting circuit (+ 12 m up the D-400 ladder) | 0.24 | 230 / 1 | 70 | 1.1 | 0.746 | 2.5 | 26.9 | 0.60 | — | minimum size |
| **F-06** | ASH-DB | HT-420 heat-tracing circuit | 0.45 | 230 / 1 | 23 | 2.0 | 0.801 | 2.5 | 28.8 | 0.37 | — | minimum size |
| **F-07** | UPS-DB (control building) | DCS and SIS cabinets (in the building rack room) | 0.72 | 230 / 1 | 15 | 3.5 | 0.801 | 2.5 | 28.8 | 0.39 | — | minimum size |

F-01 (P-400) is the upstream unit's cable, sized here to show the starting-drop check of a motor; the unit itself has no motor.

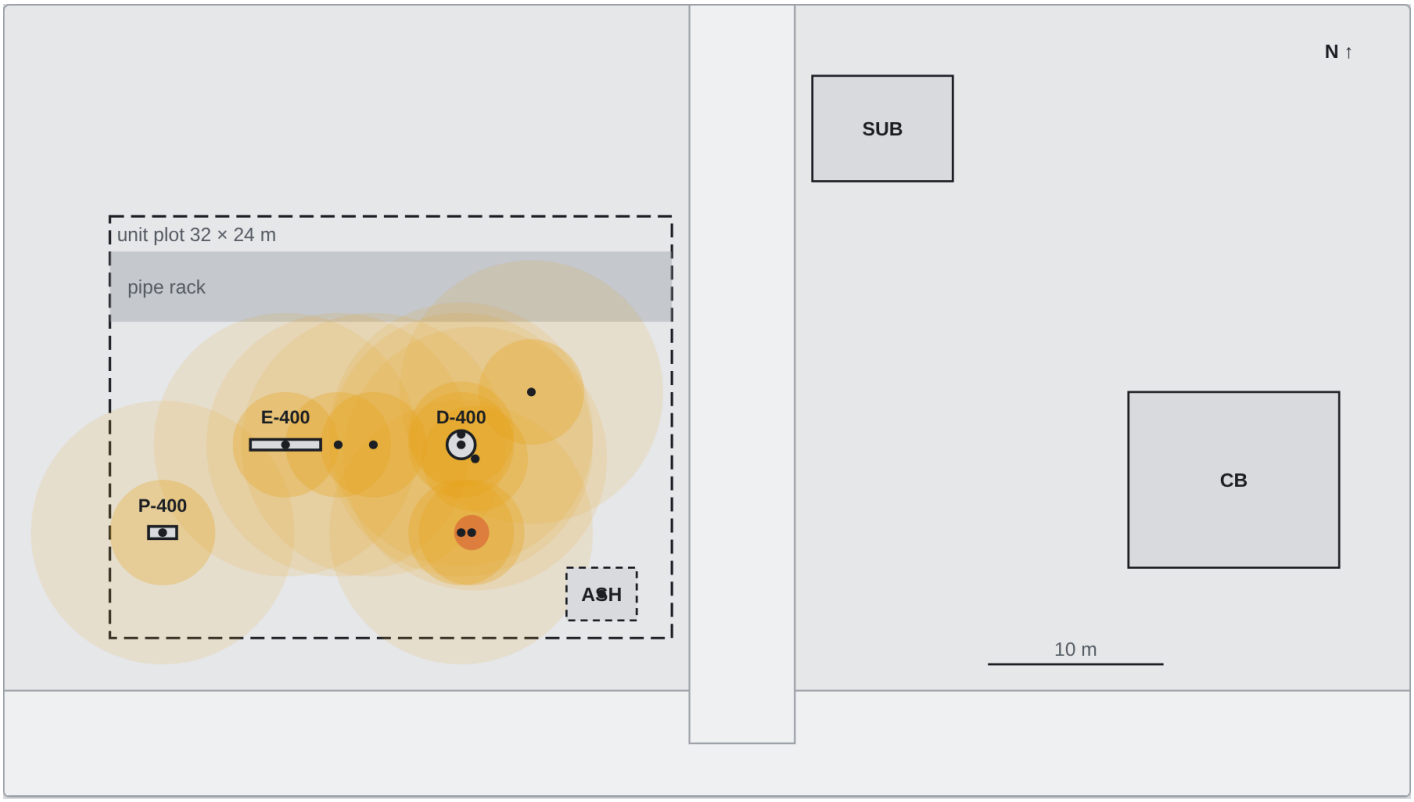
## 5 · Hazardous-area classification

| Fluid | AIT used °C | Source | MESG mm | Group | LFL–UFL % | Flash point | Vap./air |
|---|---|---|---|---|---|---|---|
| n-butane | 287 | 287 °C (NFPA 497); 365–372 °C in other tables — the lower is used | 1.0 | IIA | 1.8–8.4 | −60 °C (gas) | 2.0 |
| n-hexane | 225 | 225 °C (NFPA 497); 233 °C (IEC 60079-20-1) — the lower is used | 0.9 | IIA | 1.2–7.4 | −22 °C | 3.0 |
| D-400 vapour (76.8 wt% butane) | 225 | taken as the lowest component, n-hexane (conservative) | 0.9 | IIA | 1.6–8.2 | gas | 2.2 |
| Hot oil (mineral heat-transfer oil, typical) | 330 | ≈ 330 °C (typical mineral oil, assumed) | — | — | — | ≈ 210 °C (typical, assumed): handled at 180 °C, below its flash point | — |
| Nitrogen | — | inert | — | — | — | — | 1.0 |

Lowest AIT 225 °C (n-hexane) → equipment temperature class **T3** (max. surface 200 °C), gas group **IIA**. Hot surfaces: E-400 shell (hot oil supply 180 °C) 180 °C; Hot-oil design temperature 205 °C; Heat-tracing sheath, self-regulating (max. withstand, typical) 85 °C; Hot feed after E-400 (TIC-401 74 °C; design 205 °C) 205 °C — all below 225 °C.

| Source | Description | x, y m | Elev. m | Grade | Zone | R m |
|---|---|---|---|---|---|---|
| S01 | D-400 nozzles, flanges, bridle and instrument connections | 20.0, 11.0 | 3.0–8.6 | secondary | 2 | 3.0 |
| S02 | E-400 channel and tube-side flanges (feed at 600 kPa g) | 10.0, 11.0 | 1.2 | secondary | 2 | 3.0 |
| S03 | FCV-400 let-down station (stem packing, flanges, HV-401/402/403) | 15.0, 11.0 | 1.0 | secondary | 2 | 3.0 |
| S04 | XV-408 feed SIS valve | 13.0, 11.0 | 1.0 | secondary | 2 | 3.0 |
| S05 | LCV-400 bottoms station and XV-418 | 20.0, 6.0 | 0.8 | secondary | 2 | 3.0 |
| S06 | PCV-420 vapour station on the top platform | 24.0, 14.0 | 9.0 | secondary | 2 | 3.0 |
| S07 | PSV-400 inlet / outlet flanges (discharge closed to the flare) | 20.0, 11.6 | 9.2 | secondary | 2 | 3.0 |
| S08 | Drain HV-415 to the closed drain (drain hub, any pit or trench) | 20.8, 10.2 | 0 | secondary; pits below grade Zone 1 | 2 | 3.0 |

| Source | Description | x, y m | Elev. m | Grade | Zone | R m |
|---|---|---|---|---|---|---|
| S09 | AT-410 sample take-off and grab-sample point | 20.6, 6.0 | 0.8 | primary (sampling) | 1 | 1.0 |
| S14 | P-400 pump seal and flanges (upstream scope at the battery limit; classified by the upstream unit) | 3.0, 6.0 | 0.5 | secondary | 2 | 3.0 |
| S10 | Analyser shelter interior (sample handling, vents to a safe place) | 28.0, 2.5 | 0–3 | secondary, ventilated | 2 | inside |
| S11 | TCV-401, XV-409 and hot-oil flanges (oil below its flash point) | 10.0, 14.0 | 1.0 | none | — | — |
| S12 | PCV-430 nitrogen station (inert) | 23.0, 8.0 | 1.0 | none | — | — |
| S13 | Vapour and flare headers on the rack (all-welded, no flanges in the unit) | 16.0, 20.0 | 5.0 | none | — | — |

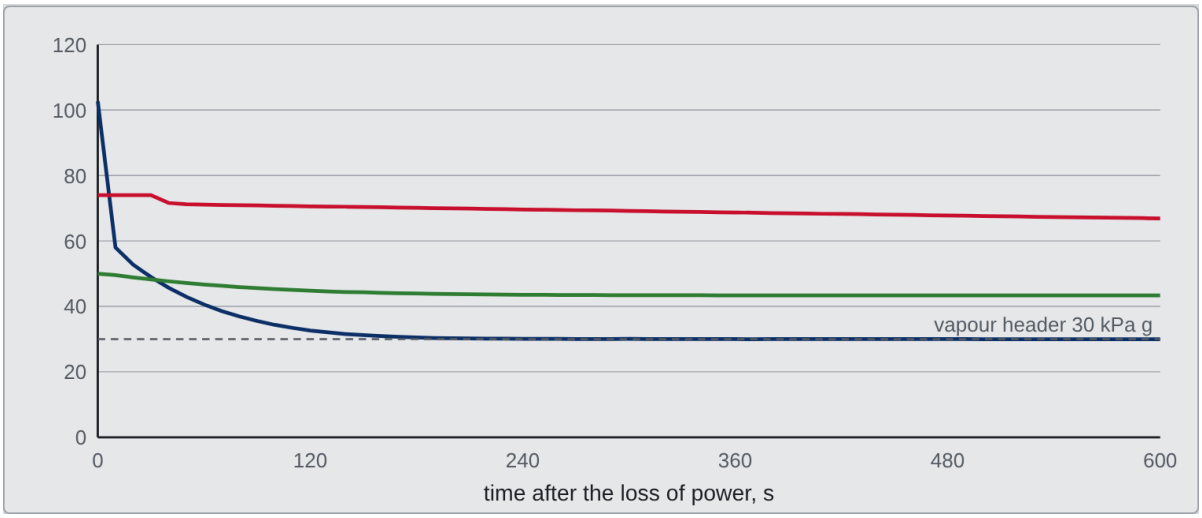



| Building | Nearest source m | Outside the zones |
|---|---|---|
| Control building (operators) | 34.0 | yes |
| Electrical substation | 20.0 | yes |
| Analyser shelter (assumed) | 5.8 | **no** (EE-01) |

## 6 · Loss of power

| Element | Service | Fail position | On loss of power | Result for the unit |
|---|---|---|---|---|
| **FCV-400** | Feed let-down valve | FC | positioner loses its signal (and air): spring closes | no feed into D-400 |
| **TCV-401** | Hot-oil 3-way valve | to bypass | spring drives the plug to the bypass port | no hot oil through E-400 |
| **LCV-400** | Bottoms valve | FC | spring closes | liquid kept in D-400: no gas blow-by to the receiver |
| **PCV-420** | Vapour valve | FO | spring opens | D-400 relieves to the vapour header and settles near its 30 kPa g |
| **XV-408 / 409 / 418** | SIS valves | FC | solenoids XY de-energise, actuator vents, spring closes | feed, hot oil and bottoms isolated (de-energise to trip) |
| **PCV-430** | $N_2$ regulator | — | self-acting, needs no power | keeps the blanket at 30 kPa g |
| **PSV-400** | Relief valve | — | spring-loaded, needs no power | protects D-400 whatever happens |
| **LG-400** | Level gauge | — | magnetic, needs no power | the level stays readable in the field |
| **LSHH-416, PSHH-426, LSLL-417** | SIS switches | contact opens | a dead input reads as a trip | I-1 / I-2 go to the safe state, they do not hide a trip |
| **AT-400 / 410 / 420** | Analysers | — | not on the UPS: stop; restart and recalibrate | spec readings lost until restarted |
| **HT-400 / 420** | Heat tracing | — | off: sample lines cool below the dew point | analyser readings wrong after a restart until the lines are warm |
| **DCS, SIS** | Control and safety systems | — | on the UPS for 30 min (assumed), then outputs to 0 mA / de-energised | the unit goes to the fail-safe state above |
| **P-400** | Feed pump (upstream) | — | stops if its own supply is lost | no feed pressure (FIC-400 low alarm) |

Model run on a copy of the simulator at design: all XVs closed, DCS outputs at the fail positions (FCV-400 0 %, TCV-401 0 %, LCV-400 0 %, PCV-420 100 %), PCV-430 still working.




| t s | PI-420 kPa g | LI-400 % | FI-400 m³/h | FI-420 Nm³/h | FI-410 m³/h | TI-401 °C | FI-405 t/h | FCV-400 % | TCV-401 % | LCV-400 % | PCV-420 % |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | 102.7 | 50.1 | 16.0 | 610.6 | 14.0 | 74.0 | 11.0 | 73.5 | 30.5 | 63.4 | 38.6 |
| 60.0 | 40.6 | 46.8 | 0.0 | 1,582.8 | 0.0 | 71.1 | 0.0 | 0.0 | 0.0 | 0.0 | 100.0 |
| 120.0 | 32.7 | 44.9 | 0.0 | 775.8 | 0.0 | 70.6 | 0.0 | 0.0 | 0.0 | 0.0 | 100.0 |
| 240.0 | 30.1 | 43.7 | 0.0 | 125.1 | 0.0 | 69.6 | 0.0 | 0.0 | 0.0 | 0.0 | 100.0 |
| 360.0 | 30.0 | 43.5 | 0.0 | 0.0 | 0.0 | 68.7 | 0.0 | 0.0 | 0.0 | 0.0 | 100.0 |
| 480.0 | 30.0 | 43.5 | 0.0 | 0.0 | 0.0 | 67.8 | 0.0 | 0.0 | 0.0 | 0.0 | 100.0 |
| 600.0 | 29.9 | 43.5 | 0.0 | 0.0 | 0.0 | 66.9 | 0.0 | 0.0 | 0.0 | 0.0 | 100.0 |

## 7 · Engineering queries

| No. | Subject | Query | Why it matters | Proposed answer | Status |
|---|---|---|---|---|---|
| EE-01 | Analyser shelter not on the plot plan | The load list and the zone plan place the analyser shelter at x 26–30 m, y 1–4 m (assumed); it is not in core/plot.json. There it is 5.8 m from the nearest source: inside the 7.5 m ground-level Zone 2 band. | Shelter location sets the sample-line lengths (tracing loads, analyser lag) and the Ex rating of the shelter equipment. | Layout to fix the shelter on the plot plan; if it stays in Zone 2, pressurise the shelter (or use Ex ec / Ex d equipment) and add a gas detector that trips the shelter power. | OPEN |
| EE-02 | P-400 keeps running against a closed XV-408 | I-1 closes XV-408 but P-400 (upstream unit, 7.5 kW) is not stopped: it runs at shut-off (650 kPa g) until someone acts. | A centrifugal pump at shut-off heats its contents; the butane-rich feed can flash in the casing and damage the seal. | Ask the upstream unit for a minimum-flow line on P-400, or add a hard-wired stop of P-400 from the SIS (DO to the upstream MCC) with a run-status DI back. | OPEN |
| EE-03 | SIS UPS autonomy not specified | DCS and SIS take 0.72 kW on the UPS; the documents give no autonomy. 30 min is assumed here (UPS 2 kVA, battery ≈ 0.5 kWh). | After the autonomy the SIS de-energises: the unit trips to the safe state (by design), but the operator loses all indication. | Fix the autonomy (time to bring the unit to a safe, monitored state by hand); consider a dedicated SIS UPS or a separate SIS distribution from the redundant UPS. | OPEN |
| EE-04 | AT-400 sample line also needs tracing | EP-09 proposes tracing for the AT-420 sample line only. AT-400 samples the same saturated vapour (49.2 °C) and condenses at ambient 28 °C in the same way. | Condensate in the $N_2$ analyser sample reads low $N_2$; the start-up sweep (AI-400 < 2 %) would be judged on a wrong value. | Trace AT-400 sample line as HT-400 (load list) or share one traced sample line for AT-400 and AT-420. | OPEN |
| EE-05 | No gas detection in the unit | The P&IDs show no flammable-gas detectors. The unit has Zone 1 at the sample point and pits, a Zone 2 band at grade and an analyser shelter inside it. | Detection gives the warning that the zone classification assumes (secondary releases are found and stopped quickly). | Add point gas detectors at the LCV-400 / sample area, at E-400 and in the analyser shelter (trip of the shelter power); count them in the I/O. | OPEN |